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B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

CS 19 202 0404 AUTOMATA LANGUAGES AND COMPUTATIONS

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1 Design a minimized Deterministic Finite Automata.
- CO2 Analyse and generate regular expressions for any structure.
- CO3 Demonstrate that a given language is regular or not.
- CO4 Design new context free grammar.
- CO5 Design Push Down Automata for any context free grammar.
- CO6 Analyse and design turing machines for any problem.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A
(Answer *ALL* questions)

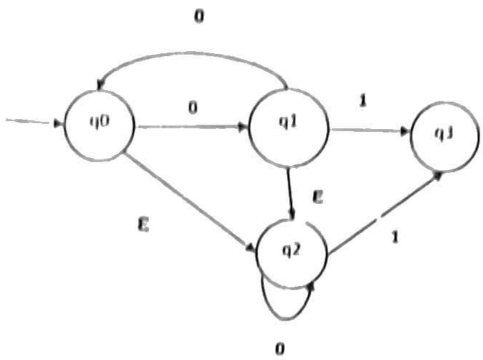
		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	Compare and contrast Non deterministic and Deterministic Finite Automata based on their definitions. As an example, give a transition diagram for each.	3	L1	1	1,5
	(b)	Compare and contrast Moore and Mealy machines. Draw a transition diagram for each machine.	3	L1	1	1,5
	(c)	Write regular expressions for (i) binary strings having 00 as a substring. (ii) binary strings starting with '01' and ending with '10'. (iii) strings made of a, b, c which contain 'abc' as a substring.	3	L3	2	1,2
	(d)	Write a regular grammar to accept any binary string starting with '0'.	3	L3	3	1,2
	(e)	Remove left recursion from the grammar rule $S \rightarrow Sab \mid a \mid b$.	3	L2	4	3,5
	(f)	Explain the working of a Push Down Automata.	3	L1	5	2,3
	(g)	Explain any one method used to construct a Turing machine.	3	L2	6	5
	(h)	A universal turing machine is a programmable turing machine. Justify the statement.	3	L1	6	5

PART B

	(4 × 12 = 48)				
II.	Construct NFA to accept all binary strings starting with '0' and ending with '1'. Note that the length of all binary strings is greater than or equal to 3. Then convert the NFA to a DFA.	12	L5	1	1,5

OR

			Marks	BL	CO	PO
III.		Eliminate epsilon transitions from the following NFA. Here q3 is the final state.	12	L3	1	1,2, 3,9



			Marks	BL	CO	PO
IV.	(a)	Using pumping lemma for regular languages show that $L = \{a^n b^n / n>0\}$ is not regular.	6	L3	2	1,3,4
	(b)	Convert the following regular grammar to finite automata. $S \rightarrow aB \mid bX \mid cA$ $A \rightarrow aB \mid b$ $B \rightarrow aX$ $X \rightarrow a$	6	L3	2	1,3,4
		OR				
V.		Design NFA for the following regular expressions. (i) 1*0 (0 + 11)*110* (ii) ab*a (ab + ba)*a (iii) 11* (00*0 + 11*1) 00.	12	L3	1	1,5
VI.		Design a PDA to accept binary strings which are palindromes and the character in the middle of the binary string is 'C'. Use the empty stack condition for acceptance.	12	L3	5	4,5
		OR				
VII.	(a)	What are the three methods to simplify context free grammars? Simplify the following CFG. $S \rightarrow aAbX \mid aX \mid aAAb$ $A \rightarrow ad \mid bSX \mid a$ $B \rightarrow aB \mid a$ $C \rightarrow aB \mid d$ $X \rightarrow a$	6	L3	4	1
	(b)	Convert the following grammar to chomsky normal form. $S \rightarrow aAc \mid Xa \mid Baad$ $X \rightarrow a$ $A \rightarrow ad \mid bSA \mid ab$ $B \rightarrow aB$	6	L3	4	2,3
VIII.		Design a turing machine to accept the language $L = \{a^n b^{n+1} / n>=1\}$.	12	L6	6	5
		OR				
IX.	(a)	Explain the following (i) Multi track turing machines. (ii) Chomsky hierarchy. (iii) Halting problem of turing machines.	6	L1	6	5
	(b)	Define a turing machine and explain how its works.	6	L2	5	1,2

Bloom's Taxonomy Levels
L1 – 15%, L2 – 10%, L3 – 55%, L5 – 10%, L6 – 10%.

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